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U. S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

10 CFR 50.73

SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-388 (387)/2017-007-00
UNIT 2 LICENSE NO. NPF-22
UNIT 1 LICENSE NO. NPF-14
PLA-7641

Docket No. 50-388
50-387

Attached is Licensee Event Report (LER) 50-388(387)/2017-007-00. The LER reports an event involving a loss of secondary containment differential pressure due to changes in supply air flow. This event was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(v)(C) as a condition that could have prevented fulfillment of a safety function.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new regulatory commitments.

A handwritten signature in black ink, appearing to be "B. Berryman", with a stylized flourish at the end.

B. Berryman

Attachment: LER 50-388(387)/2017-007-00

Copy: NRC Region I
Ms. T. E. Hood, NRC Project Manager
Ms. L. H. Micewski, NRC Sr. Resident Inspector
Mr. M. Shields, PA DEP/BRP



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME Susquehanna Steam Electric Station Unit 2	2. DOCKET NUMBER 05000388	3. PAGE 1 OF 3
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4. TITLE Secondary Containment Declared Inoperable Due to Supply Air Flow

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
08	08	2017	2017	- 007	- 00			2017	Susquehanna Steam Electric Station Unit 1	05000387
									FACILITY NAME	DOCKET NUMBER
										05000

9. OPERATING MODE 1	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
10. POWER LEVEL 100	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
☐ 50.73(a)(2)(i)(C) ☐ OTHER Specify in Abstract below or in NRC Form 366A				

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

Nicole Pagliaro – Licensing Specialist - Nuclear Regulatory Affairs

TELEPHONE NUMBER (Include Area Code)

(570) 542-6578

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX
X	VA	FAN	N/A	Y					

14. SUPPLEMENTAL REPORT EXPECTED☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE)☒ NO**15. EXPECTED SUBMISSION DATE**

MONTH	DAY	YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On August 8, 2017 at approximately 2043 hours, Secondary Containment Zone 2 experienced a loss of differential pressure following the replacement of the Zone 2 ventilation supply side filters.

Technical Specification Surveillance Requirement (SR) 3.6.4.1.1 was not met. Zone 2 differential pressure was recovered following the slight adjustment of the Reactor Building Zone 2 inlet dampers at approximately 2052 hours. Zones 1 and 3 were not affected. This event is being reported under 10 CFR 50.73(a)(2)(v)(C) as a condition that could have prevented fulfillment of a safety function.

The cause was determined to be increased supply air flow following the replacement of the supply side filters. An action to revise future work instructions to ensure operations support is available to adjust Zone 2 differential pressure control following Zone 2 Supply side filter replacement.

There were no actual consequences to the health and safety of the public as a result of this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Susquehanna Steam Electric Station Unit 2	05000388	YEAR 2017	SEQUENTIAL NUMBER - 007	REV NO. - 00

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power

Unit 2 – Mode 1, approximately 100 percent Rated Thermal Power

There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event.

EVENT DESCRIPTION

On August 8, 2017 at approximately 2043 hours, Secondary Containment Zone 2 [EIS Code: VA] experienced a loss of differential pressure (dP) following replacement of the Zone 2 ventilation supply side filters. All fans were running; however, the 'B' discharge damper controller output was found to be outside of its normal operating band.

The following is a timeline of events:

8/8/17: The filters on Zone 2 ventilation supply side were replaced during day shift.

8/8/17 2030: The Control Room received Reactor Building (RB) [EIS Code: NG] FAN ROOM Elevation 779' 2C275 PANEL trouble ALARM.

8/8/17 2043: Operations reported the local alarm was for RB hi-lo DP and RB Zone 2 dP is 0.15" vacuum water gauge. All fans are running, but noted that PDD-27578B (2V205B Discharge Damper) is indicating dual indication.

8/8/17 2044: Entered Technical Specification 3.6.4.1 for Unit 1 and Unit 2 due to Secondary Containment inoperable with Zone 2 dP <0.25" vacuum water gauge.

8/8/17 2052: Operations restored building dP to >0.25" vacuum water gauge after slight adjustment to the Zone 2 inlet dampers and swapping Zone 2 Exhaust Fans ("A" in-service, "B" inStandby).

8/8/17 2112: Exited condition for Technical Specification 3.6.4.1 for both units.

As a result, Technical Specification Surveillance Requirement (SR) 3.6.4.1.1 was not met. Zone 2 differential pressure was recovered to > 0.25" vacuum water gauge following slight adjustment of the Reactor Building Zone 2 inlet dampers at approximately 2052 hours. Zones 1 and 3 were not affected.

CAUSE OF EVENT

The cause was determined to be increased supply air flow following the replacement of the supply side filters and less than adequate written communication in the work instructions on the impact of filter replacements on reactor building differential pressure.

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CONTINUATION SHEET**

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ANALYSIS/SAFETY SIGNIFICANCE

An engineering evaluation was performed and concluded that secondary containment could have performed its safety function of isolating as assumed in the accident analysis and also of re-establishing 0.25 inches vacuum (drawdown) within the assumed accident analysis time (10 minutes). Therefore, the subject event did not cause a loss of safety function. This event will not be counted as a safety system functional failure (SSFF) for the NRC performance indicator based on the engineering analysis that shows there was no loss of ability to fulfill the safety function.

CORRECTIVE ACTIONS

Corrective actions include revising the Preventative Maintenance work instructions to ensure operations support is available to monitor output of the discharge damper controller and to adjust the inlet damper as needed during filter replacement.

PREVIOUS SIMILAR EVENTS

The following are recent LERs involving loss of secondary containment due to fan issues:

LER 50-388 / 2017-006-00 "LER for Loss of Secondary Containment Zone II differential pressure- ENS for Loss of Safety Function," dated September 6, 2017.

LER 50-388 / 2017-005-00 "Loss of Secondary Containment due to trip of 2V217A Zone II Filtered Exhaust fan," dated August 18, 2017.

LER 50-387 / 2017-004-00 "LER for Loss of Secondary Containment Zone 3 differential pressure- ENS for loss of Safety Function," dated August 4, 2017.

LER 50-387 / 2017-003-00 "Loss of Secondary Containment Zone 3 due to fan trip," dated May 5, 2017

LER 50-387 / 2016-003-00 "Unit 2 Zone 3 HVAC unable to maintain Zone 3 dP >0.25"VG," dated May 3, 2016

LER 50-387 / 2015-013-00, "Loss of Secondary Containment due to failure of running fans," dated February 2, 2016.